

Project: EcoTrack - Environmental Monitoring & Sustainability Management System

1. Introduction

EcoTrack is a web-based platform designed for municipalities, environmental agencies, NGOs, and industries to monitor environmental parameters, track sustainability initiatives, and ensure compliance with regulations. It enables citizens to report environmental issues, agencies to monitor air/water quality, industries to log emissions, and administrators to generate sustainability reports.

The system supports workflows for environmental data collection, citizen reporting, compliance monitoring, sustainability project tracking, and analytics. It ensures transparency and accountability by maintaining audit trails, dashboards, and performance metrics across environmental governance.

Actors / Users

- **Citizen:** Reports environmental issues, accesses sustainability dashboards.
- **Agency Officer:** Monitors environmental parameters, validates citizen reports.
- **Industry Representative:** Logs emissions, submits compliance documents.
- **Environmental Scientist:** Analyzes data, provides recommendations.
- **Administrator:** Configures workflows, manages users, oversees reporting.
- **Compliance Officer:** Ensures adherence to environmental policies, audits records.

2. Module Overview

- 2.1 Identity & Access Management
- 2.2 Citizen Reporting & Issue Management
- 2.3 Environmental Data Collection & Monitoring
- 2.4 Industry Emissions & Compliance Logging
- 2.5 Sustainability Project & Initiative Tracking
- 2.6 Compliance & Audit Management
- 2.7 Reporting & Analytics
- 2.8 Notifications & Alerts

3. Architecture Overview

- **Frontend:** Angular or React for responsive dashboards.
- **Backend:** REST API-based microservices (Identity, Reporting, Monitoring, Industry, Compliance, Projects, Reports).
- **Database:** Relational DB (MySQL/PostgreSQL/SQL Server).

- **Deployment:** Cloud/on-prem with API gateway, WAF, IoT sensor integration, centralized logging.

4. Module Wise Design

4.1 Identity & Access Management

Features:

Provides secure authentication, role-based access control, and audit logging.

Entities:

- User(UserID, Name, Role [Citizen/Agency/Industry/Scientist/Admin/Compliance], Email, Phone, Status)
- AuditLog(AuditID, UserID, Action, Resource, Timestamp)

4.2 Citizen Reporting & Issue Management

Features:

Citizens report environmental issues, agencies validate and track resolution.

Entities:

- Issue(IssueID, CitizenID, Type [Pollution/Waste/Deforestation], Location, Date, Status)
- Resolution(ResolutionID, IssueID, OfficerID, Actions, Date, Status)

4.3 Environmental Data Collection & Monitoring

Features:

Collects data from IoT sensors, monitors air/water quality, tracks trends.

Entities:

- Sensor(SensorID, Location, Type [Air/Water/Noise], Status)
- SensorData(DataID, SensorID, ParametersJSON, Timestamp)
- Analysis(AnalysisID, DataID, ScientistID, Findings, Date, Status)

4.4 Industry Emissions & Compliance Logging

Features:

Industries log emissions, submit compliance documents, track violations.

Entities:

- EmissionLog(LogID, IndustryID, Type [CO2/NOx/SOx], Quantity, Date, Status)
- IndustryDocument(DocumentID, IndustryID, DocType [Permit/Compliance], FileURI, UploadedDate, VerificationStatus)

4.5 Sustainability Project & Initiative Tracking

Features:

Tracks sustainability projects, monitors progress, evaluates impact.

Entities:

- Project(ProjectID, Title, Description, StartDate, EndDate, Budget, Status)
- Milestone(MilestoneID, ProjectID, Title, Date, Status)
- Impact(ImpactID, ProjectID, MetricsJSON, Date, Status)

4.6 Compliance & Audit Management

Features:

Ensures adherence to environmental policies, audits industry and agency records.

Entities:

- ComplianceRecord(ComplianceID, EntityID, Type [Emission/Project/Policy], Result, Date, Notes)
- Audit(AuditID, OfficerID, Scope, Findings, Date, Status)

4.7 Reporting & Analytics

Features:

Generates dashboards and reports on environmental parameters, compliance, and sustainability metrics.

Entities:

- Report(ReportID, Scope [Issue/Emission/Project], Metrics, GeneratedDate)

4.8 Notifications & Alerts

Features:

Sends alerts for environmental thresholds, compliance deadlines, project milestones.

Entities:

- Notification(NotificationID, UserID, EntityID, Message, Category [Issue/Emission/Project/Compliance], Status, CreatedDate)

5. Deployment Strategy

- **Local:** Angular/React frontend, Spring Boot/.NET Core backend, local DB.
- **Production:** Cloud/on-prem deployment, API gateway, WAF, IoT sensor integration, centralized logging.

6. Database Design

Tables:

- User(UserID, Name, Role, Email, Phone, Status)
- AuditLog(AuditID, UserID, Action, Resource, Timestamp)
- Issue(IssueID, CitizenID, Type, Location, Date, Status)
- Resolution(ResolutionID, IssueID, OfficerID, Actions, Date, Status)
- Sensor(SensorID, Location, Type, Status)
- SensorData(DataID, SensorID, ParametersJSON, Timestamp)
- Analysis(AnalysisID, DataID, ScientistID, Findings, Date, Status)
- EmissionLog(LogID, IndustryID, Type, Quantity, Date, Status)
- IndustryDocument(DocumentID, IndustryID, DocType, FileURI, UploadedDate, VerificationStatus)
- Project(ProjectID, Title, Description, StartDate, EndDate, Budget, Status)
- Milestone(MilestoneID, ProjectID, Title, Date, Status)
- Impact(ImpactID, ProjectID, MetricsJSON, Date, Status)
- ComplianceRecord(ComplianceID, EntityID, Type, Result, Date, Notes)
- Audit(AuditID, OfficerID, Scope, Findings, Date, Status)
- Report(ReportID, Scope, Metrics, GeneratedDate)
- Notification(NotificationID, UserID, EntityID, Message, Category, Status, CreatedDate)

7. User Interface Design

- **Citizen Portal:** Issue reporting, sustainability dashboards, alerts.
- **Agency Dashboard:** Issue validation, sensor monitoring, compliance tracking.
- **Industry Console:** Emission logging, compliance document submission.
- **Scientist Workbench:** Data analysis, recommendations, research insights.
- **Admin Panel:** User management, workflow setup, reporting.
- **Compliance Officer Console:** Policy monitoring, audits, compliance reporting.

8. Non-Functional Requirements

- **Performance:** Handle 50,000 concurrent users across city networks.
- **Security:** Role-based access, encrypted data storage, immutable audit logs.
- **Scalability:** Support multi-city, multi-agency rollout.
- **Availability:** 99.9% uptime.
- **Maintainability:** Modular microservices, API versioning, automated migrations.
- **Observability:** Centralized logging, KPIs (pollution reduction, compliance adherence, project impact).

9. Assumptions & Constraints

- Initial rollout for a single city before multi-city expansion.
- IoT sensor integration limited to pilot areas in Phase 1.
- Notifications limited to in-app and SMS/email alerts.
- Fully implementable using Java Spring Boot/ASP.NET Core, Angular/React, IoT integration, and relational DB.